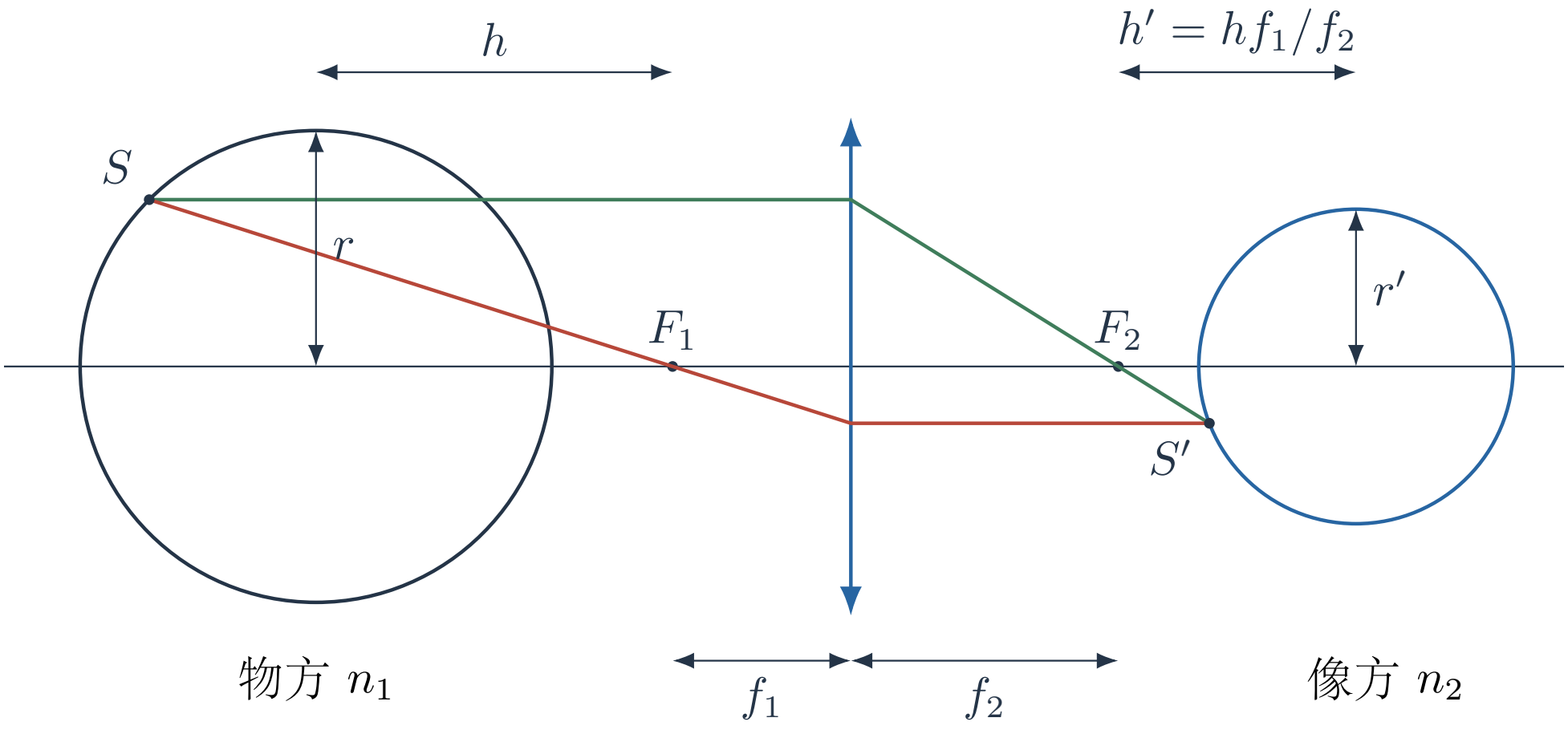

$$x_1 x_2 = f_1 f_2, \quad y_2/y_1 = -f_1/x_1$$